Year 2 Pure Mathematics

Chapter 4

Binomial expansion

Practice Booklet

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4.1 Expanding $(1+x)^n$

Practice Book, printed page 29.

1. Find the binomial expansion up to and including the term in x^3 of:

a. $\dfrac{1}{(1+x)^2}$

b. $(1+x)^{\frac{1}{2}}$

c. $(1+x)^{\frac{4}{3}}$

d. $(1+x)^{-\frac{1}{5}}$

e. $(1+x)^{-5}$

f. $(1+x)^{-\frac{5}{2}}$

Hint

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots + \left(\frac{n(n-1)\ldots(n-r+1)}{r!}\right)x^r + \cdots$$

This expansion is valid when $|x| < 1$, $n \in \mathbb{R}$.

2. For each expression:

i. find the binomial expansion up to and including the term in x^3

ii. state the range of values of x for which the expansion is valid.

a. $\dfrac{1}{(1+2x)^3}$

b. $\sqrt{1-x}$

c. $\left(1+\dfrac{x}{2}\right)^{\frac{2}{3}}$

d. $(1-5x)^{-4}$

e. $\dfrac{1}{\sqrt[3]{1-2x}}$

f. $\left(1-\dfrac{x}{3}\right)^{-2}$

Hint

Replace x with (bx) in the binomial expansion formula. Remember that $(bx)^n = b^n x^n$.

The expansion of $(1+bx)^n$ is valid for $|bx| < 1$, or $|x| < \dfrac{1}{|b|}$

3.

a. Find the first four terms, in ascending powers of x, in the binomial expansion of $\sqrt{1-4x}$.

b. By substituting $x = 0.01$ in your expansion, find a decimal approximation for $\sqrt{6}$.

Hint

When $x = 0.01$,

$\sqrt{1-4x} = \sqrt{0.96} = \sqrt{6 \times \frac{16}{100}} = \frac{2}{5}\sqrt{6}$

c. State why the approximation is valid for $x = 0.01$.

4. $f(x) = \dfrac{4+x}{\sqrt{1+3x}}$

Hint

Write $f(x)$ as $(4+x)(1+3x)^{-\frac{1}{2}}$

a. Find the first three terms, in ascending powers of x, of the binomial expansion of $f(x)$. Give each term in its simplest form.

b. State the range of values of x for which the expansion is valid.

5. E/P Given that the binomial expansion of $(1+kx)^{-5}$, $|kx| < 1$, is $1 - 8x + Ax^2 + \ldots$

a. find the value of the constant k (2 marks)

b. find the value of the constant A, giving your answer as a fraction in its simplest form. (3 marks)

6. E/P $f(x) = \dfrac{1-x}{1+3x}$

a. Show that the series expansion of $f(x)$ up to and including the term in x^3 is $1 - 4x + 12x^2 - 36x^3$ (4 marks)

b. State the range of values of x for which the expansion is valid. (1 mark)

7. E/P $g(x) = \dfrac{5}{1+4x} - \dfrac{3}{1-2x}$

a. Find the series expansion of $g(x)$ in ascending powers of x, up to and including the term in x^2. Simplify each term. (6 marks)

b. State the range of values of x for which the expansion of $g(x)$ is valid. (1 mark)

c. Find the percentage error made in using your expansion in part a to estimate the value of $g(0.01)$. Give your answer to 2 significant figures. (3 marks)

8. E/P

a. Find the binomial expansion of $\sqrt{1-6x}$, $|x| < \dfrac{1}{6}$, in ascending powers of x up to and including the term in x^3, simplifying each term. (4 marks)

b. By substituting $x = \dfrac{1}{300}$, find a decimal approximation for $\sqrt{2}$. Give your answer to 5 decimal places. (5 marks)

9. E/P

a. Use the binomial expansion to show that $\sqrt{\dfrac{1-5x}{1+x}} \approx 1 - 3x - \dfrac{3x^2}{2}$, $|x| < 1$. (6 marks)

b. Substitute $x = \dfrac{1}{8}$ into $\sqrt{\dfrac{1-5x}{1+x}} \approx 1 - 3x - \dfrac{3x^2}{2}$ to obtain an approximation for $\sqrt{3}$. Give your answer in the form $\dfrac{a}{b}$ where a and b are integers. (3 marks)

4.2 Expanding $(a+bx)^n$

Practice Book, printed page 30.

1. For each expression:

i. find the binomial expansion up to and including the term in x^3

ii. state the range of values of x for which the expansion is valid.

a. $\dfrac{1}{(2+3x)^2}$

b. $\sqrt{4-x}$

c. $(8+2x)^{\frac{2}{3}}$

Hint

To expand $(a+bx)^n$ where a and b are constants, first take out a factor of a^n.

$$(a+bx)^n = \left(a\left(1+\frac{b}{a}x\right)\right)^n = a^n\left(1+\frac{b}{a}x\right)^n$$

The expansion of $(a+bx)^n$ is valid for $\left|\frac{b}{a}x\right| < 1$ or $|x| < \left|\frac{a}{b}\right|$.

2. For each expression:

i. find the binomial expansion up to and including the term in x^3

ii. state the range of values of x for which the expansion is valid.

a. $\dfrac{3}{5-2x}$

b. $\dfrac{1-x}{(2-x)^2}$

c. $\sqrt{\dfrac{4+x}{1-x}}$

Hint

$$\sqrt{\frac{4+x}{1-x}} = (4+x)^{\frac{1}{2}}(1-x)^{-\frac{1}{2}}$$
$$= 2\left(1+\frac{x}{4}\right)^{\frac{1}{2}}(1-x)^{-\frac{1}{2}}$$

3. E/P Given $\mathrm{f}(x) = (5-3x)^{-3}$,

a. find the binomial expansion of $\mathrm{f}(x)$, in ascending powers of x, up to and including the term in x^3. Give each coefficient as a simplified fraction. (5 marks)

b. State the range of values of x for which the expansion is valid. (1 mark)

4. E/P

a. Find the first three terms in the binomial expansion of $\dfrac{1}{\sqrt{6+2x}}$, $|x| < 3$.
Give each coefficient in simplified surd form. (5 marks)

b. Hence or otherwise, find the first three terms in ascending powers of x, of the expansion of $\dfrac{3x-1}{\sqrt{6+2x}}$ (4 marks)

5. E/P The first three terms in the binomial expansion of $\dfrac{1}{\sqrt{a+bx}}$ are $\dfrac{1}{3} - \dfrac{2x}{27} + \dfrac{2x^2}{81}$.

a. Find the values of the constants a and b. (6 marks)

b. State the range of values of x for which the expansion is valid. (1 mark)

c. Find the coefficient of the term in x^3 in its simplest form. (2 marks)

6. E/P

a. Use the binomial expansion to expand $(8-5x)^{\frac{1}{3}}$, $|x| < \dfrac{8}{5}$, in ascending powers of x, up to and including the term in x^2, giving each term as a simplified fraction. (5 marks)

b. Use your expansion, with a suitable value of x, to obtain an approximation for $\sqrt[3]{7}$.
Give your answer to 3 decimal places. (2 marks)

7. E/P $g(x) = \dfrac{4}{2-4x} - \dfrac{2}{3-5x}$

a. Find the series expansion of $g(x)$ in ascending powers of x, up to and including the term in x^2. Simplify each term. (6 marks)

b. State the range of values of x for which the expansion of $g(x)$ is valid. (1 mark)

c. Find the exact value of $g(0.01)$. Round your answer to 7 decimal places. (2 marks)

d. Find the percentage error made in using your expansion from part a to estimate the value of $g(0.01)$. Give your answer to 2 significant figures. (3 marks)

8. E/P

a. Use the binomial expansion to expand $\sqrt[3]{\dfrac{1+x}{8-3x}}$ in ascending powers of x, up to and including the term in x^2, giving each term as a simplified fraction. (8 marks)

b. State the range of values of x for which the expansion is valid. (1 mark)

c. By substituting $x = 0.1$ into your expansion, find an approximation for $\sqrt[3]{7}$, giving your answer to 5 decimal places. (3 marks)

4.3 Using partial fractions

Practice Book, printed page 31.

1.

a. Express $\dfrac{4x+1}{(1+x)(2-x)}$ as partial fractions.

Hint

Partial fractions can be used to simplify more complicated algebraic fractions and make them easier to expand using the binomial expansion. ← Section 1.3

b. Hence, or otherwise, expand $\dfrac{4x+1}{(1+x)(2-x)}$ in ascending powers of x, up to and including the term in x^2.

c. State the range of values of x for which the expansion is valid.

Hint

Choose a range of values for x that is valid for all the binomial expansions that you use.

2.

a. Express $\dfrac{3x}{(3+x)^2}$ as partial fractions.

Hint

As $(3+x)$ is a repeated linear factor, use two partial fractions with denominators $(3+x)$ and $(3+x)^2$ respectively. ← Section 1.4

b. Use the binomial expansion to show that $\dfrac{3x}{(3+x)^2} \approx \dfrac{1}{3}x + Bx^2 + Cx^3$, where B and C are constants to be determined.

c. State the range of values of x for which the expansion is valid.

3.

a. Express $\dfrac{2x^2+5x-8}{(x-5)(x+4)}$ using partial fractions.

Hint

As the degree of the numerator is the same as the degree of the denominator, first divide the numerator by the denominator. ← Section 1.5

b. Hence, or otherwise, expand $\dfrac{2x^2+5x-8}{(x-5)(x+4)}$ in ascending powers of x, up to and including the

term in x^2.

c. State the range of values for x for which the expansion is valid.

4. E/P

a. Express $f(x) = \dfrac{8x-1}{(1-2x)(1+4x)}$ as partial fractions. (3 marks)

b. Hence, or otherwise, expand $\dfrac{8x-1}{(1-2x)(1+4x)}$ in ascending powers of x, up to and including the term in x^2. (6 marks)

c. State the range of values for x for which the expansion is valid. (1 mark)

5. E/P $\dfrac{10x+4}{(1+2x)^2} \equiv \dfrac{A}{1+2x} + \dfrac{B}{(1+2x)^2}$, $|x| < \dfrac{1}{2}$, where A and B are constants.

a. Find the values of A and B. (3 marks)

b. Hence, or otherwise, find the first three terms in the expansion of $\dfrac{10x+4}{(1+2x)^2}$. (6 marks)

6. E/P $\dfrac{3x^2+5x-7}{(x+4)(x-3)} \equiv A + \dfrac{B}{x+4} + \dfrac{C}{x-3}$, where A, B and C are constants.

a. Find the values of A, B and C. (4 marks)

b. Hence, or otherwise, find the first three terms in the expansion of $\dfrac{3x^2+5x-7}{(x+4)(x-3)}$ (6 marks)

c. State the range of values for x for which the expansion is valid. (1 mark)

7. E/P $f(x) = \dfrac{2x^2+5x+12}{(3x-1)^2(x-2)} \equiv \dfrac{A}{3x-1} + \dfrac{B}{(3x-1)^2} + \dfrac{C}{x-2}$

a. Find the values of A, B and C. (4 marks)

b. Hence, or otherwise, find the series expansion of $f(x)$, in ascending powers of x, up to and including the term in x^2. Simplify each term. (6 marks)

c. Explain why the expansion is not valid for $x = 0.5$. (1 mark)

Problem solving: Set A

Bronze

The binomial expansion of $(1+10x)^{\frac{1}{2}}$ in ascending powers of x, up to and including the term in x^3, is $1+5x+px^2+qx^3$, $|10x|<1$.

a. Find the value of p and the value of q. (4 marks)

b. Use this expansion with your values of p and q together with an appropriate value of x to obtain an estimate of $(1.5)^{\frac{1}{2}}$. (3 marks)

c. Find the percentage error made in using your expansion to estimate the value of $(1.5)^{\frac{1}{2}}$.
Give your answer to 2 significant figures. (3 marks)

Silver

a. Find the binomial expansion of $\sqrt{1-6x}$ in ascending powers of x, up to and including the term in x^3, simplifying each term. (4 marks)

b. Substitute $x=\dfrac{1}{9}$ into the binomial expansion to find an approximation for $\sqrt{3}$.
Give your answer as a fraction in its simplest form. (5 marks)

c. Explain why the expansion is valid for this value of x. (1 mark)

Gold

a. Use binomial expansions to show that $\sqrt{\dfrac{1+3x}{1-x}} \approx 1+2x+2x^3$. (6 marks)

A student attempts to use this expansion to find an approximation for $\sqrt{5}$. The first few steps of the student's working are shown below.

$$\frac{1+3x}{1-x}=5$$
$$1+3x=5-5x$$
$$x=\frac{1}{2}$$
Substitute $x=\dfrac{1}{2}$ into expansion

b. Explain the error in the student's working. (2 marks)

c. By substituting a suitable value of x into the approximation, obtain an approximation for $\sqrt{5}$, giving your answer to 5 decimal places. You must state the value of x you chose and justify your choice. (1 mark)

Problem solving: Set B

Bronze

a. Use the binomial expansion to show that $\dfrac{1}{2-3x} \approx \dfrac{1}{2}+\dfrac{3x}{4}+\dfrac{9x^2}{8}$ and state the range of values of x for which the expansion is valid. (5 marks)

b. Obtain the binomial expansion of $\dfrac{1}{(1-x)^2}$ up to and including the term in x^2. (2 marks)

c. Given that $\dfrac{3x^2-4}{(2-3x)(1-x)^2}$ can be written in the form $\dfrac{A}{(2-3x)}+\dfrac{B}{(1-x)}+\dfrac{C}{(1-x)^2}$, find the values of the constants A, B and C. (5 marks)

d. Hence find the binomial expansion of $\dfrac{3x^2-4}{(2-3x)(1-x)^2}$ up to and including the term in x^2. (3 marks)

e. State the range of values of x for which this expansion is valid. (1 mark)

Silver

a. Express $f(x)=\dfrac{8x^2+12x+4}{x^2+x-6}$ using partial fractions. (5 marks)

b. Hence, or otherwise, find the series expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. Simplify each term. (5 marks)

c. State the range of values of x for which the expansion is valid. (1 mark)

Gold

By considering the series expansion of $\frac{1}{2x^2+5x+3}$, or otherwise, use binomial expansions to show that $\frac{\sqrt{1+3x}}{2x^2+5x+3} \approx \frac{1}{3} - \frac{x}{18} - \frac{109x^2}{216}$ (10 marks)

Now try this ⟶ Exam question bank Q14, Q33, Q51, Q57, Q60, Q70, Q81

Selected exam question bank questions

Question 14

a. Use the binomial expansion, in ascending powers of x, to show that $\sqrt{9+x} = 3 + \frac{1}{6}x + kx^2 + \ldots$ where k is a rational constant to be found. (4 marks)

A student attempts to substitute $x = 1$ into both sides of this equation to find an approximate value for $\sqrt{10}$.

b. State, giving a reason, if the expansion is valid for this value of x. (1 mark)

Question 33

a. Expand $\frac{1}{\sqrt{4-3x}}$, where $|x| < \frac{4}{3}$, in ascending powers of x, up to and including the term in x^2. Simplify each term. (5 marks)

b. Hence, or otherwise, find the first 3 terms in the expansion of $\frac{x+8}{\sqrt{4-3x}}$ as a series in ascending powers of x. (4 marks)

Question 51

a. Show that the binomial expansion of $(9+7x)^{\frac{1}{2}}$ in ascending powers of x up to and including the term in x^2 is $3 + \frac{7}{6}x + kx^2$, giving the value of the constant k as a simplified fraction. (4 marks)

b.

i. Use the expansion from part a, with $x = \frac{9}{25}$, to find an approximate value for $\sqrt{2}$.
Give your answer in the form $\frac{p}{q}$, where p and q are integers.

ii. Explain why substituting $x = \frac{9}{25}$ into this binomial expansion leads to a valid approximation. (4 marks)

Question 57

a. Use binomial expansions to show that $\sqrt{\frac{1+6x}{1-x}} \approx 1 + \frac{7x}{2} - \frac{21x^2}{8}$ (6 marks)

A student substitutes $x = \frac{1}{5}$ into both sides of the approximation shown in part a in an attempt to find an approximation to $\sqrt{11}$.

b. Give a reason why the student should not use $x = \frac{1}{5}$ (1 mark)

c. Substitute $x = \frac{2}{65}$ into $\sqrt{\frac{1+6x}{1-x}} \approx 1 + \frac{7x}{2} - \frac{21x^2}{8}$ to obtain an approximation to $\sqrt{11}$.
Give your answer to 5 decimal places. (3 marks)

Question 60

a. Expand $\frac{1}{(3-4x)^2}$, $|x| < \frac{3}{4}$, in ascending powers of x up to and including the term in x^2, giving each term as a simplified fraction. (5 marks)

Given that the binomial expansion of $\frac{2+kx}{(3-4x)^2}$, $|x| < \frac{3}{4}$, is $\frac{2}{9} + \frac{7x}{27} + Ax^2 + \ldots$

b. find the value of the constant k (2 marks)

c. find the value of the constant A. (2 marks)

Question 70

$$f(x) = \frac{5}{3+4x} - \frac{3}{2-5x},\ |x| < \frac{2}{5}$$

a. Show that the first three terms in the series expansion of $f(x)$ can be written as $\frac{1}{6} - \frac{215x}{36} - \frac{1385x^2}{216}$ (7 marks)

b. Find the exact value of f(0.01). Round your answer to 7 decimal places. (2 marks)

c. Find the percentage error made in using the series expansion in part a to estimate the value of f(0.01). Give your answer to 2 significant figures. (3 marks)

Question 81

$$f(x) = \frac{36x^2 + 15x - 14}{(3x-2)^2(2-x)},\ |x| < \frac{2}{3}$$

Given that $f(x)$ can be expressed in the form $\frac{A}{3x-2} + \frac{B}{(3x-2)^2} + \frac{C}{2-x}$,

a. find the values of the constants A, B and C. (4 marks)

b. Hence, or otherwise, find the series expansion of $f(x)$ in ascending powers of x, up to and including the term in x^2. Simplify each term. (7 marks)

c. Find the percentage error made in using the series expansion in part b to estimate the value of f(0.1). Give your answer to 2 significant figures. (4 marks)

Answers

4.1 Expanding $(1+x)^n$

1.

a. $1 - 2x + 3x^2 - 4x^3$

b. $1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16}$

c. $1 + \frac{4x}{3} + \frac{2x^2}{9} - \frac{4x^3}{81}$

d. $1 - \frac{x}{5} + \frac{3x^2}{25} - \frac{11x^3}{125}$

e. $1 - 5x + 15x^2 - 35x^3$

f. $1 - \frac{5x}{2} + \frac{35x^2}{8} - \frac{105x^3}{16}$

2.

a.

i. $1 - 6x + 24x^2 - 80x^3$

ii. $|x| < \frac{1}{2}$

b.

i. $1 - \frac{x}{2} - \frac{x^2}{8} - \frac{x^3}{16}$

ii. $|x| < 1$

c.

i. $1 + \frac{x}{3} - \frac{x^2}{36} + \frac{x^3}{162}$

ii. $|x| < 2$

d.

i. $1 + 20x + 250x^2 + 2500x^3$

ii. $|x| < \frac{1}{5}$

e.

i. $1 + \frac{2x}{3} + \frac{8x^2}{9} + \frac{112x^3}{81}$

ii. $|x| < \frac{1}{2}$

f.

i. $1 + \frac{2x}{3} + \frac{x^2}{3} + \frac{4x^3}{27}$

ii. $|x| < 3$

3.

a. $1 - 2x - 2x^2 - 4x^3$

b. 2.44949

c. $\sqrt{1-4x}$ is valid for $|x| < \frac{1}{4}$ and $0.01 < \frac{1}{4}$

4.

a. $4 - 5x + 12x^2$

b. $|x| < \frac{1}{3}$

5.

a. $\frac{8}{5}$

b. $\frac{192}{5}$

6.

a. $(1+3x)^{-1} \approx 1 - 3x + 9x^2 - 27x^3$
So $(1-x)(1-3x+9x^2-27x^3)$
$= 1 - 3x + 9x^2 - 27x^3 - x + 3x^2 - 9x^3$
$= 1 - 4x + 12x^2 - 36x^3$

b. $|x| < \frac{1}{3}$

7.

a. $2 - 26x + 68x^2$

b. $|x| < \frac{1}{4}$

c. 0.019%

8.

a. $1 - 3x - \frac{9x^2}{2} - \frac{27x^3}{2}$

b. $\frac{7}{10}\sqrt{2} \approx 0.9899495$ gives $\sqrt{2} \approx 1.41421$

9.

a. $(1-5x)^{\frac{1}{2}} \approx 1 - \frac{5x}{2} - \frac{25x^2}{8}$
and $(1+x)^{-\frac{1}{2}} \approx 1 - \frac{x}{2} + \frac{3x^2}{8}$
$\left(1 - \frac{5x}{2} - \frac{25x^2}{8}\right)\left(1 - \frac{x}{2} + \frac{3x^2}{8}\right)$
$\approx 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x}{2} + \frac{5x^2}{4} - \frac{25x^2}{8} = 1 - 3x - \frac{3x^2}{2}$

b. $\frac{128}{77}$

4.2 Expanding $(a+bx)^n$

1.

a.

i. $\frac{1}{4} - \frac{3x}{4} + \frac{27x^2}{16} - \frac{27x^3}{8}$

ii. $|x| < \frac{2}{3}$

b.

i. $2 - \frac{x}{4} - \frac{x^2}{64} - \frac{x^3}{512}$

ii. $|x| < 4$

c.

i. $4 + \frac{2x}{3} - \frac{x^2}{36} + \frac{x^3}{324}$

ii. $|x| < 4$

2.

a.

i. $\frac{3}{5} + \frac{6x}{25} + \frac{12x^2}{125} + \frac{24x^3}{625}$

ii. $|x| < \frac{5}{2}$

b.

i. $\frac{1}{4} - \frac{x^2}{16} - \frac{x^3}{16}$

ii. $|x| < 2$

c.

i. $2 + \frac{5x}{4} + \frac{55x^2}{64} + \frac{365x^3}{512}$

ii. $|x| < 1$

3.

a. $\frac{1}{125} + \frac{9x}{625} + \frac{54x^2}{3125} + \frac{54x^3}{3125}$

b. $|x| < \frac{5}{3}$

4.

a. $\frac{1}{\sqrt{6}} - \frac{x}{6\sqrt{6}} + \frac{x^2}{24\sqrt{6}}$

b. $-\frac{1}{\sqrt{6}} + \frac{19x}{6\sqrt{6}} - \frac{13x^2}{24\sqrt{6}}$

5.

a. $a = 9, b = 4$

b. $|x| < \frac{9}{4}$

c. $-\frac{20}{2187}$

6.

a. $2 - \frac{5x}{12} - \frac{25x^2}{288}$

b. $x = 0.2$: $\sqrt[3]{7} \approx 1.913$

7.

a. $\frac{4}{3} + \frac{26x}{9} + \frac{166x^2}{27}$

b. $|x| < \frac{1}{2}$

c. 1.3628502

d. 0.00097%

8.

a. $\frac{1}{2} + \frac{11x}{48} - \frac{11x^2}{576}$

b. $|x| < 1$

c. 1.91305

4.3 Using partial fractions

1.

a. $\frac{4x+1}{(1+x)(2-x)} \equiv \frac{-1}{1+x} + \frac{3}{2-x}$

b. $\frac{1}{2} + \frac{7x}{4} - \frac{5x^2}{8}$

c. $|x| < 1$

2.

a. $\frac{3x}{(3+x)^2} \equiv \frac{3}{3+x} - \frac{9}{(3+x)^2}$

b. $\frac{x}{3} - \frac{2x^2}{9} + \frac{x^3}{9}$

c. $|x| < 3$

3.

a. $2 + \frac{67}{9(x-5)} - \frac{4}{9(x+4)}$

b. $\frac{2}{5} - \frac{27x}{100} - \frac{133x^2}{2000}$

c. $|x| < 4$

4.

a. $\frac{1}{1-2x} - \frac{2}{1+4x}$

b. $-1 + 10x - 28x^2$

c. $|x| < \frac{1}{4}$

5.

a. $A = 5, B = -1$

b. $4 - 6x + 8x^2$

6.

a. $A = 3, B = -3, C = 5$

b. $\frac{7}{12} - \frac{53x}{144} - \frac{401x^2}{1728}$

c. $|x| < 3$

7.

a. $A = -\frac{44}{15}, B = -\frac{25}{3}, C = \frac{6}{5}$

b. $-6 - \frac{83x}{2} - \frac{795x^2}{4}$

c. Expansion of f(x) is valid for $|x| < \frac{1}{3}$

As $0.5 > \frac{1}{3}$, the expansion is not valid for $x = 0.5$

Problem solving: Set A

Bronze

a. $p = -\frac{25}{2}, q = \frac{125}{2}$

b. $x = 0.05$: $(1.5)^{\frac{1}{2}} \approx 1.2265625$

c. 0.15%

Silver

a. $1 - 3x - \frac{9x^2}{2} - \frac{27x^3}{2}$

b. $\sqrt{3} \approx \frac{27}{16}$

c. Expansion is valid for $|x| < \frac{1}{6}$ and $\frac{1}{9} < \frac{1}{6}$

Gold

a. $(1+3x)^{\frac{1}{2}} \approx 1 + \frac{3x}{2} - \frac{9x^2}{8} + \frac{27x^3}{16}$

$(1-x)^{-\frac{1}{2}} \approx 1 + \frac{x}{2} + \frac{3x^2}{8} + \frac{5x^3}{16}$

Multiplying these together gives $1 + 2x + 2x^3 + \ldots$

b. The expansion is valid for $|x| < \frac{1}{3}$ and $\frac{1}{2} > \frac{1}{3}$

c. To ensure $|x| < \frac{1}{3}$, equate the argument to e.g. $\frac{5}{4}$, which gives $x = \frac{1}{17}$. Then $\sqrt{5} \approx 2.23611$ to 5 d.p.

Problem solving: Set B

Bronze

a. $\dfrac{1}{2-3x} = (2-3x)^{-1} = 2^{-1}\left(1-\dfrac{3x}{2}\right)^{-1}$

$= \dfrac{1}{2}\left(1+(-1)\times\left(\dfrac{3x}{2}\right)+\dfrac{(-1)(-2)}{2!}\times\left(\dfrac{3x}{2}\right)^2+\ldots\right)$

So $\dfrac{1}{2-3x} \approx \dfrac{1}{2}+\dfrac{3x}{4}+\dfrac{9x^2}{8}$, valid for $|x| < \dfrac{2}{3}$

b. $1+2x+3x^2$

c. $A=-24, B=9, C=1$

d. $-2-7x-15x^2$

e. $|x| < \dfrac{2}{3}$

Silver

a. $\mathrm{f}(x) \equiv 8-\dfrac{8}{x+3}+\dfrac{12}{x-2}$

b. $-\dfrac{2}{3}-\dfrac{19x}{9}-\dfrac{97x^2}{54}$

c. $|x| < 2$

Gold

$\dfrac{1}{2x^2+5x+3} \equiv \dfrac{1}{x+1}-\dfrac{2}{2x+3}$

$\dfrac{\sqrt{1+3x}}{2x^2+5x+3} = \dfrac{\sqrt{1+3x}}{x+1}-\dfrac{2\sqrt{1+3x}}{2x+3}$

$\dfrac{1}{x+1} \approx 1-x+x^2$ and $-\dfrac{2}{2x+3} \approx -\dfrac{2}{3}+\dfrac{4x}{9}-\dfrac{8x^2}{27}$

$\sqrt{1+3x} = (1+3x)^{\frac{1}{2}} \approx 1+\dfrac{3x}{2}-\dfrac{9x^2}{8}$

$\dfrac{\sqrt{1+3x}}{x+1} \approx (1-x+x^2)\left(1+\dfrac{3x}{2}-\dfrac{9x^2}{8}\right) \approx 1+\dfrac{x}{2}-\dfrac{13x^2}{8}$

$-\dfrac{2\sqrt{1+3x}}{2x+3} \approx \left(1+\dfrac{3x}{2}-\dfrac{9x^2}{8}\right)\left(-\dfrac{2}{3}+\dfrac{4x}{9}-\dfrac{8x^2}{27}\right)$

$\approx -\dfrac{2}{3}-\dfrac{5x}{9}+\dfrac{121x^2}{108}$

$1+\dfrac{x}{2}-\dfrac{13x^2}{8}+\left(-\dfrac{2}{3}-\dfrac{5x}{9}+\dfrac{121x^2}{108}\right) = \dfrac{1}{3}-\dfrac{x}{18}-\dfrac{109x^2}{216}$

Selected exam question bank answers

Question 14

a. $\sqrt{9+x} = (9+x)^{\frac{1}{2}} = 9^{\frac{1}{2}}\left(1+\dfrac{x}{9}\right)^{\frac{1}{2}}$

$= 3\left(1+\dfrac{1}{2}\times\dfrac{x}{9}+\dfrac{1}{2}\times\left(-\dfrac{1}{2}\right)\times\dfrac{x^2}{81}\times\dfrac{1}{2!}+\ldots\right)$

$= 3+\dfrac{x}{6}-\dfrac{x^2}{216}+\ldots$ so $k=-\dfrac{1}{216}$

b. Expansion is valid for $|x| < 9$, so it is valid for $x=1$.

Question 33

a. $\frac{1}{2} + \frac{3x}{16} + \frac{27x^2}{256}$

b. $4 + 2x + \frac{33x^2}{32}$

Question 51

a. $(9+7x)^{\frac{1}{2}} = 9^{\frac{1}{2}}\left(1+\frac{7x}{9}\right)^{\frac{1}{2}}$

$= 3\left(1 + \frac{1}{2}\times\frac{7}{9}x + \frac{1}{2}\times\left(-\frac{1}{2}\right)\times\left(\frac{7x}{9}\right)^2\times\frac{1}{2!}+\ldots\right)$

$= 3 + \frac{7}{6}x - \frac{49}{216}x^2$

b.

i. $\sqrt{2} \approx \frac{5651}{4000}$

ii. Expansion is valid for $|x| < \frac{9}{7}$ and $\frac{9}{25} < \frac{9}{7}$

Question 57

a. $\sqrt{1+6x} \approx 1 + 3x - \frac{9}{2}x^2$ and $(1-x)^{-\frac{1}{2}} \approx 1 + \frac{x}{2} + \frac{3x^2}{8}$

$\left(1+3x-\frac{9}{2}x^2\right)\left(1+\frac{x}{2}+\frac{3x^2}{8}\right) = 1 + \frac{7x}{2} - \frac{21x^2}{8}$

b. Expansion is valid for $|x| < \frac{1}{6}$ and $\frac{1}{5} > \frac{1}{6}$

c. 3.31562

Question 60

a. $\frac{1}{9} + \frac{8x}{27} + \frac{16x^2}{27}$

b. -3

c. $\frac{8}{27}$

Question 70

a. $5(3+4x)^{-1} - 3(2-5x)^{-1}$

$= 5\left(\frac{1}{3} - \frac{4x}{9} + \frac{16x^2}{27}\right) - 3\left(\frac{1}{2} - \frac{5x}{4} + \frac{25x^2}{8}\right)$

$= \frac{1}{6} - \frac{215x}{36} - \frac{1385x^2}{216}$

b. 0.1062753

c. 0.026%

Question 81

a. $A = 18, B = 9, C = 10$

b. $-\frac{7}{4} - \frac{17}{4}x - \frac{61}{16}x^2$

c. 0.10%